

Next-Token Prediction

Tokens, embeddings, and one complete neural language-model calculation

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ES 667 · Deep Learning · IIT Gandhinagar

L11 classified spatial sites; L12 classifies the next time site

PUBLIC L11 · SEGMENTATION

one categorical distribution at
every spatial location
output support $H \times W$



PUBLIC L12 · LANGUAGE

one categorical distribution at the
next ordered location
output support V

PUBLIC L11 · SEGMENTATION

one categorical distribution at
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PUBLIC L12 · LANGUAGE

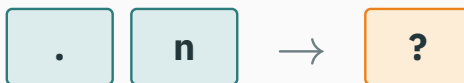
one categorical distribution at the
next ordered location
output support V

The classifier and cross-entropy stay. Order, causality, and feeding predictions back are new.

Commit before the model reveals anything

QUESTION

HELD CASE · OPENING COMMITMENT Vocabulary [., a, e, i, n, v]; visible context .n; predict the next token.



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OUTPUT

What six numbers must the model return?

CHOICE

Will greedy and sampling always pick the same token?

LEARNING

Which embedding rows can this example update?

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Commit silently. We will derive every answer from one set of weights and revisit all three.

The 80-minute route protects one numerical spine

CORE · 55 MIN

boundary + causal
windows → embedding
lookup → fixed-context
MLP → stable softmax →
CE → $p - y$ → token NLL /
PPL

SHOULD · 15 MIN

autoregressive loop →
greedy / sampling →
temperature → suffix-
blind diagnostic →
opening revisit

OPTIONAL · 10 MIN

full gradient matrices →
learned representation →
Colab and primary
provenance

The final RNN handoff is CORE even when optional material is skipped.

Three clocks reuse the distribution but change the question

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

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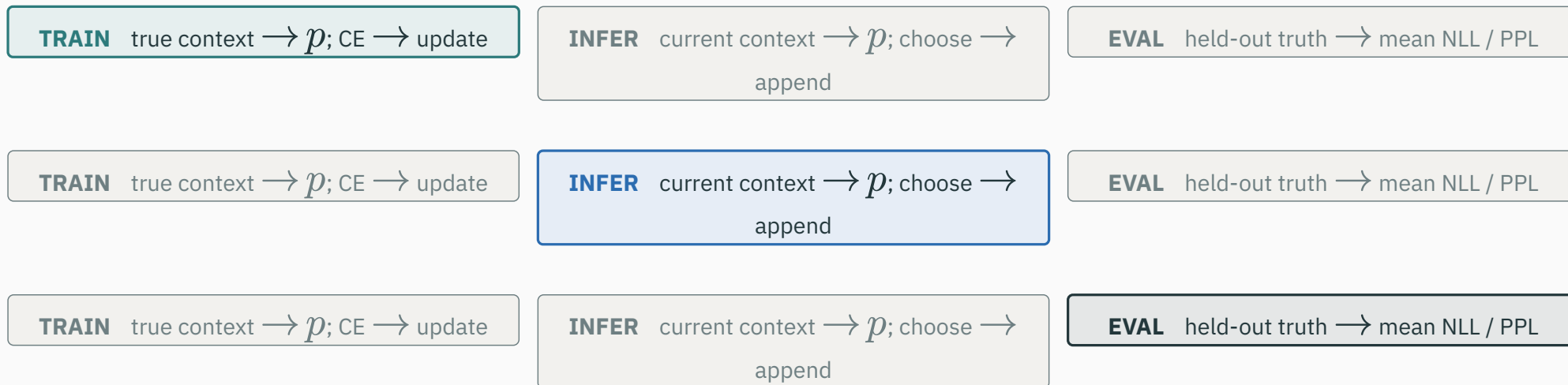
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Three clocks reuse the distribution but change the question



Always ask: are we updating weights, choosing a token, or scoring held-out truth?

By the end, one sentence should be operational

V VISUAL

A language model maps an observed prefix to a distribution over the next token.

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REPRESENT

turn categorical ids into
learned vectors without
inventing numeric order

PREDICT

produce V logits,
normalize, and train with
token cross-entropy

DIAGNOSE

separate decoding
choices from an
architecture that cannot
see enough history

— observed context / TRAIN — model score or probability / INFER — target or chosen token — correct / retained — loss or error — EVAL

One tiny corpus supplies every visible token

TRAIN NAMES

naveen · navin · neev · neena · vani
naina · avni · anvi · veena · navi

HELD-OUT NAMES

navina · naveena · veenavi · anvina

Toy convention: . is one shared boundary symbol. Repeat it $k = 2$ times as left padding; use one . target to end a name. It is not punctuation.

Tokenization chooses the units the model must predict

unit	example	vocabulary	sequence
character	learning → l e a r n i n g	small	long
word	deep learning → deep learning	very large	short
subword	unpredictable → un predict able	moderate	moderate

Characters keep every row visible. Production usually separates <BOS>, <EOS>, <PAD>, and <UNK>; this toy merges <BOS> and <EOS> as ..

The vocabulary fixes ids and output coordinates

.	a	e	i	n	v
0	1	2	3	4	5

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$V = 6$ means six logits, six probabilities, and six possible next tokens.

HELD CASE · VOCABULARY The target a is coordinate 1; context .n reads ids (0, 4).

Sliding a two-character window creates seven examples

..	.n	na	av	ve	ee	en
n	a	v	e	e	n	.

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top = visible context; bottom = next-token target

HELD CASE · TRAINING PAIRS .n → a is the second example and remains our hand calculation.

Causal training shifts the same sequence by one site

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

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append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

input	..	.n	na	av	ve	ee	en
target	n	a	v	e	e	n	.

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input	..	.n	na	av	ve	ee	en
target	n	a	v	e	e	n	.

At position t , the target token is not part of the input context.

The chain rule turns sequence probability into next-token factors

Let $B = (., .)$ denote the fixed left boundary padding and let $x_7 = .$ be the end target.

$$p(x_{1:7} \mid B) = \prod_{t=1}^7 p(x_t \mid B, x_{<t}).$$

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D DERIVATION

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$$p(x_{1:7} \mid B) = \prod_{t=1}^7 p(x_t \mid B, x_{<t}).$$

This is an exact identity. The fixed-window approximation comes next.

One ledger will cross probability, loss, and evaluation

HELD CASE · SEQUENCE LEDGER Correct targets for naveen. under contexts . ., .n, na, av, ve, ee, en.

context	. .	. n	na	av	ve	ee	en
target	n	a	v	e	e	n	.
<i>p</i>(target)	.30	derive	.70	.60	.50	.40	.80

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<i>p(target)</i>	.30	derive	.70	.60	.50	.40	.80

Six probabilities are held illustrative. The orange .n → a entry will come from the exact weights.

Checkpoint: what did the chain rule assume?

QUESTION

A

tokens are independent

B

the sequence has fixed length

C

nothing extra; it is an identity

D

future tokens are visible

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future tokens are visible

C. Each exact factor conditions on the complete preceding history and boundary.

The approximation begins only when the model discards all but the latest k tokens.

A fixed-context MLP deliberately truncates the history

$$p_{\theta}(x_t \mid B, x_{<t}) \approx p_{\theta}(x_t \mid x_{t-2:t-1}).$$

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EXACT CONDITIONAL

boundary plus every earlier token
variable-length input



TOY MODEL

only two latest tokens
fixed-size input for an MLP

This model learns local spelling patterns; it cannot recover a token that has scrolled out.

A token id is a label, not a magnitude

n has id 4 and v has id 5, but

$5 - 4 = 1$ **does not mean** $v - n = 1$.

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n has id 4 and v has id 5, but

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Feeding raw ids as scalar features invents ordering and distance. Ids are categorical addresses.

The representation must preserve identity without pretending that adjacent ids are similar.

One-hot vectors give each token an independent axis

With vocabulary order [., a, e, i, n, v]:

$$o_{.} = (1 \ 0 \ 0 \ 0 \ 0 \ 0)^T, \quad o_n = (0 \ 0 \ 0 \ 0 \ 1 \ 0)^T.$$

One-hot vectors give each token an independent axis

With vocabulary order $[\cdot, a, e, i, n, v]$:

$$o_{\cdot} = (1 \ 0 \ 0 \ 0 \ 0 \ 0)^T, \quad o_n = (0 \ 0 \ 0 \ 0 \ 1 \ 0)^T.$$

For different tokens $i \neq j$, $\|o_i - o_j\|^2 = 2$.

One-hot is faithful but sparse, V -dimensional, and has no learned similarity structure.

An embedding is a learned row selected by an id

token	feature 1	feature 2
.	1	0
a	0	1
e	1	1
i	-1	1
n	0	2
v	2	0

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$$E \in \mathbb{R}^{6 \times 2}, \quad e_i = E[i] = E^T o_i.$$

The id addresses a row; gradient descent decides what the two coordinates become.

Context .n reads exactly two rows

HELD CASE · LOOKUP ids $(0, 4)$ select rows for . and n.

$$e_{.} = E[0] = (1, 0), \quad e_n = E[4] = (0, 2).$$

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The target a affects the output loss, but its embedding row is not read on this example.

Commit: concatenate or average the two rows?

QUESTION

A • AVERAGE

$$c = \frac{e_{\cdot} + e_n}{2} = (.5, 1)$$

B • CONCATENATE

$$c = [e_{\cdot}; e_n] = (1, 0, 0, 2)$$

Commit: concatenate or average the two rows?

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B. Concatenation gives the first and second context positions different coordinate blocks.

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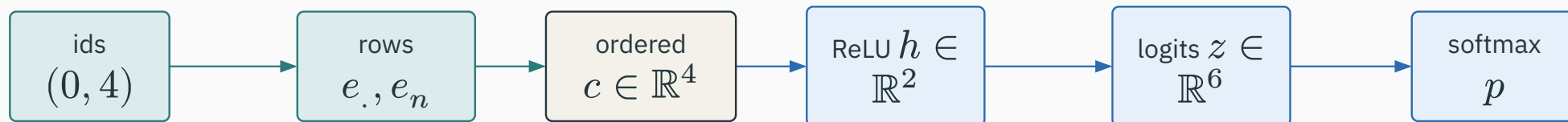
B • CONCATENATE

$$c = [e_{.}; e_n] = (1, 0, 0, 2)$$

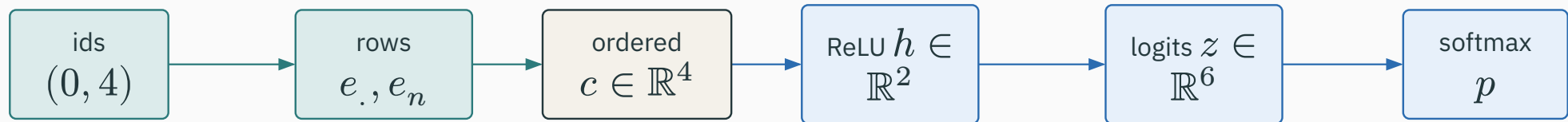
B. Concatenation gives the first and second context positions different coordinate blocks.

$$c_{.n} = (1, 0, 0, 2) \neq (0, 2, 1, 0) = c_n.$$

One position is an ordinary six-class classifier



One position is an ordinary six-class classifier



two ids $\rightarrow$ two 2-D rows $\rightarrow$ ordered 4-D context $\rightarrow$ two hidden units $\rightarrow$ six logits

Language modeling changes how examples are constructed; the local head is familiar classification.

The tensor ledger follows the pipeline

TRAIN true context $\rightarrow \underline{p}$; CE $\rightarrow$ update

INFER current context $\rightarrow \underline{p}$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$c = [E[x_{t-2}]; E[x_{t-1}]] \in \mathbb{R}^4,$$

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$$c = [E[x_{t-2}]; E[x_{t-1}]] \in \mathbb{R}^4,$$

$$u = W_h c + b_h, \quad h = \text{ReLU}(u) \in \mathbb{R}^2,$$

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$$z = W_o h + b_o \in \mathbb{R}^6, \quad p = \text{softmax}(z).$$

$W_h \in \mathbb{R}^{2 \times 4}; W_o \in \mathbb{R}^{6 \times 2}$; rows of W_o follow vocabulary order.

Count parameters once, then keep the shapes visible

block	shape	count
embedding E	$V \times d$	$Vd = 12$
hidden W_h, b_h	$H \times kd, H$	$Hkd + H =$ 10
output W_o, b_o	$V \times H, V$	$VH + V =$ 18

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$12 + 10 + 18 = 40$ parameters.

Increasing context length k grows the hidden block Hkd even before we discuss what the model can remember.

Every weight for $.n \rightarrow a$ is now explicit

HELD CASE · EXACT MODEL input $x = (., n)$; target $y = a$; $c = (1, 0, 0, 2)^T$.

$$W_h = \begin{pmatrix} 0.5 & 0 & 0 & 0.25 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \quad b_h = \begin{pmatrix} 0 \\ -1 \end{pmatrix},$$

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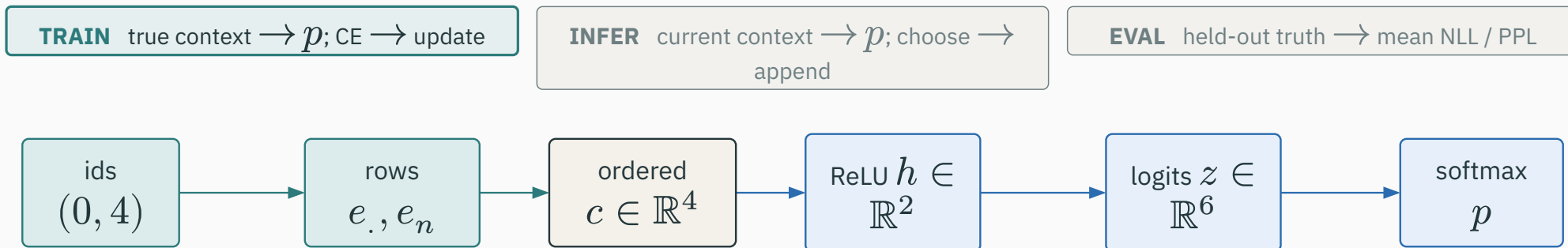
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$$W_o = \begin{pmatrix} -1 & 0 \\ 2 & 1 \\ 0 & -1 \\ -1 & 1 \\ 0 & 0 \\ 1 & -1 \end{pmatrix}, \quad b_o = 0.$$

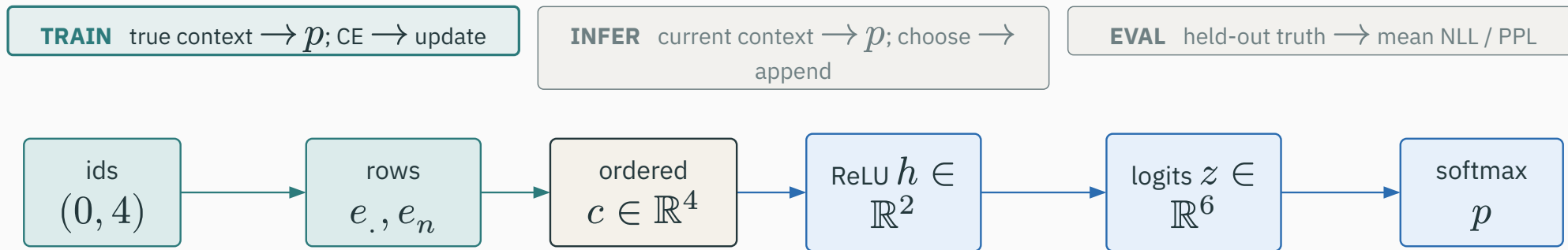
Rows of W_o are $[., a, e, i, n, v]$. Keep that order for every vector below.

Forward 1: lookup and concatenate

D DERIVATION



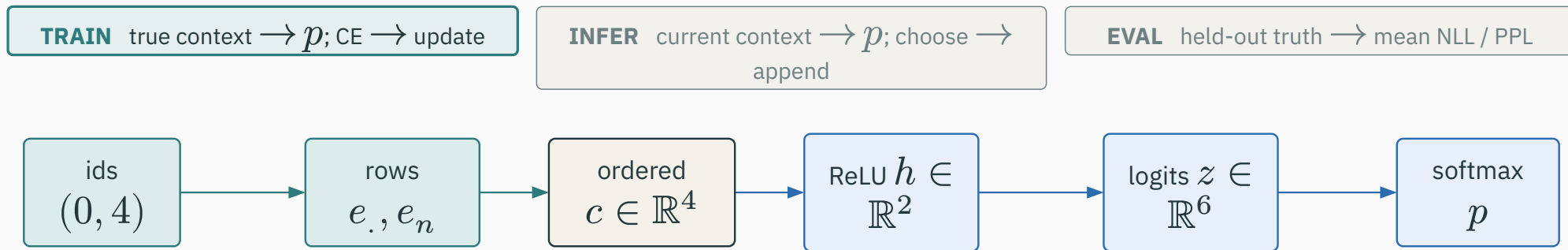
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D DERIVATION



$$E[0] = (1, 0), \quad E[4] = (0, 2),$$

$$c = [E[0]; E[4]] = (1, 0, 0, 2)^T.$$

HELD CASE · FORWARD · CONTEXT The ordered context occupies four coordinates; no averaging occurs.

Forward 2: one hidden unit turns on

$$u_1 = .5(1) + 0(0) + 0(0) + .25(2) + 0 = 1,$$

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$$h = (\max(0, 1), \max(0, -1)) = (1, 0).$$

ACTIVE

$h_1 = 1$; gradient may pass

OFF

$h_2 = 0$; ReLU blocks its gradient here

Commit: compute two logits before seeing all six

With $h = (1, 0)$, calculate the rows for target a and alternative v :

TARGET a

$$z_a = (2, 1) \cdot (1, 0) = ?$$

ALTERNATIVE v

$$z_v = (1, -1) \cdot (1, 0) = ?$$

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$$z_a = 2 \quad z_v = 1$$

A logit is an unconstrained score, not yet a probability.

Forward 3: every output row supplies one logit

D DERIVATION

HELD CASE · FORWARD · LOGITS vocabulary order [., a, e, i, n, v].

$$z = W_o h = (-1, 2, 0, -1, 0, 1)^T.$$

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$$z = W_o h = (-1, 2, 0, -1, 0, 1)^T.$$

.	a	e	i	n	v
-1	2	0	-1	0	1

Softmax will preserve this ranking while converting scores into a normalized distribution.

Stable softmax subtracts a constant before exponentiating

D DERIVATION

Naively exponentiating large logits can overflow. Softmax is unchanged by a shared shift:

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Naively exponentiating large logits can overflow. Softmax is unchanged by a shared shift:

$$\text{softmax}(z) = \text{softmax}(z - m), \quad m = \max_j z_j = 2.$$

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Naively exponentiating large logits can overflow. Softmax is unchanged by a shared shift:

$$\text{softmax}(z) = \text{softmax}(z - m), \quad m = \max_j z_j = 2.$$

$$z - m = (-3, 0, -2, -3, -2, -1).$$

Subtracting the maximum is an implementation safeguard, not a model change.

Forward 4: exponentiate and normalize

$$\exp(z - m) = (.0498, 1, .1353, .0498, .1353, .3679),$$

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$$Z = \sum_j \exp(z_j - m) = 1.7381,$$

Forward 4: exponentiate and normalize

$$\exp(z - m) = (.0498, 1, .1353, .0498, .1353, .3679),$$

$$Z = \sum_j \exp(z_j - m) = 1.7381,$$

$$p = \frac{\exp(z - m)}{Z} = (.028644, .575333, .077863, .028644, .077863, .211653).$$

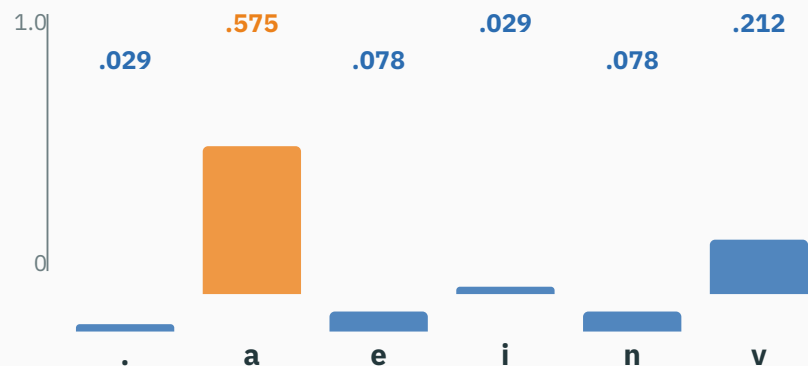
HELD CASE · FORWARD · PROBABILITY The exact orange ledger entry is now $p(a|.n) = .575333$.

The six bars share a true 0-to-1 probability axis

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

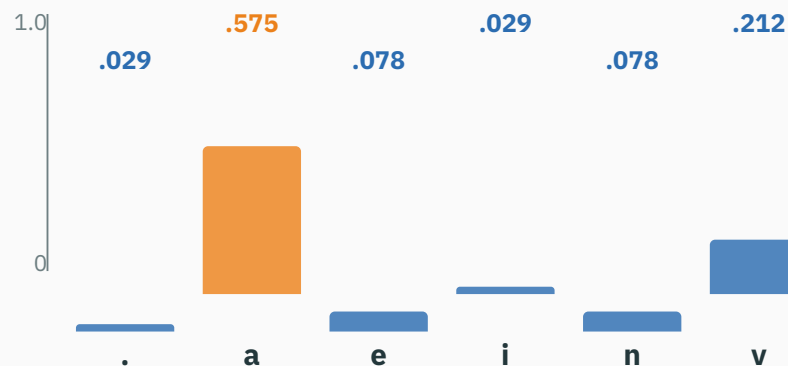


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Direct labels follow [., a, e, i, n, v]; all six probabilities sum to 1.

a is most likely, but alternatives retain nonzero probability.

Forward 5: cross-entropy reads the target coordinate

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

The target is a, so $y = (0, 1, 0, 0, 0, 0)$.

Forward 5: cross-entropy reads the target coordinate

D DERIVATION

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append

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The target is a, so $y = (0, 1, 0, 0, 0, 0)$.

$$\mathcal{L} = - \sum_j y_j \log p_j = -\log(.575333) = .552806 \text{ nats.}$$

IF p_a RISES

the loss falls

IF $p_a \rightarrow 1$

$\mathcal{L} \rightarrow 0$

The sequence ledger now contains no second version

context	..	.n	na	av	ve	ee	en
target	n	a	v	e	e	n	.
<i>p(target)</i>	.30	.575333	.70	.60	.50	.40	.80

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context	..	.n	na	av	ve	ee	en
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<i>p(target)</i>	.30	.575333	.70	.60	.50	.40	.80

$$p_{\theta(\text{naveen.}|\text{..})} \approx .30(.575333)(.70)(.60)(.50)(.40)(.80) = .0115987.$$

The sequence probability is small because all seven next-token decisions must agree.

Logs turn the same product into NLL and perplexity

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

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$$-\log(.0115987) = 4.45686 \text{ nats,}$$

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$$-\log(.0115987) = 4.45686 \text{ nats},$$

$$\text{mean NLL} = \frac{4.45686}{7} = .636694 \text{ nats/token},$$

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EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$-\log(.0115987) = 4.45686 \text{ nats},$$

$$\text{mean NLL} = \frac{4.45686}{7} = .636694 \text{ nats/token},$$

$$\text{PPL} = \exp(.636694) = 1.89022.$$

This is one illustrative name ledger, not the trained notebook's validation score.

Commit: what sign must the target-logit gradient have?

QUESTION

A • POSITIVE

gradient descent will lower z_a

B • NEGATIVE

gradient descent will raise z_a

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B. To reduce $-\log p_a$, the update must increase the target logit relative to alternatives.

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A • POSITIVE

gradient descent will lower z_a

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B. To reduce $-\log p_a$, the update must increase the target logit relative to alternatives.

Softmax plus cross-entropy makes the exact sign visible in one vector.

Backward 1: softmax plus CE gives $p - y$

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TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

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$$\nabla_z \mathcal{L} = p - y$$

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$$\nabla_z \mathcal{L} = p - y$$

$$= (.028644, -.424667, .077863, .028644, .077863, .211653).$$

- the target coordinate is negative, so gradient descent raises z_a ;
- every alternative coordinate is positive, so gradient descent lowers it.

Backward 2: output gradients are outer products

Because $z = W_o h + b_o$,

$$\nabla_{W_o} \mathcal{L} = (p - y)h^T, \quad \nabla_{b_o} \mathcal{L} = p - y.$$

Backward 2: output gradients are outer products

Because $z = W_o h + b_o$,

$$\nabla_{W_o} \mathcal{L} = (p - y)h^T, \quad \nabla_{b_o} \mathcal{L} = p - y.$$

With $h = (1, 0)$, the first column equals $p - y$ and the second column is zero.

An inactive feature emits no output-weight gradient on this example.

Backward 3: ReLU chooses which hidden signal survives

D DERIVATION

$$\nabla_h \mathcal{L} = W_o^T(p-y) = (-.694969, -.685539)^T.$$

Backward 3: ReLU chooses which hidden signal survives

D DERIVATION

$$\nabla_h \mathcal{L} = W_o^T(p-y) = (-.694969, -.685539)^T.$$

The forward pre-activation was $u = (1, -1)$, so the ReLU mask is $(1, 0)$.

Backward 3: ReLU chooses which hidden signal survives

$$\nabla_h \mathcal{L} = W_o^{T(p-y)} = (-.694969, -.685539)^T.$$

The forward pre-activation was $u = (1, -1)$, so the ReLU mask is $(1, 0)$.

$$\nabla_u \mathcal{L} = \nabla_h \mathcal{L} \odot (1, 0) = (-.694969, 0)^T.$$

The second unit had a nonzero upstream signal, but its negative pre-activation blocks that signal.

Backward 4: return to the ordered context

Since $u = W_h c + b_h$,

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$$\nabla_c \mathcal{L} = W_h^T \nabla_u \mathcal{L} = (-.347485, 0, 0, -.173742)^T.$$

Backward 4: return to the ordered context

Since $u = W_h c + b_h$,

$$\nabla_c \mathcal{L} = W_h^T \nabla_u \mathcal{L} = (-.347485, 0, 0, -.173742)^T.$$

block	$e.[1]$	$e.[2]$	$e_n[1]$	$e_n[2]$
∇_c	$-.347485$	0	0	$-.173742$

Concatenation makes the split back into two embedding rows unambiguous.

$$\nabla_{e_{\cdot}} \mathcal{L} = (-.347485, 0), \quad \nabla_{e_n} \mathcal{L} = (0, -.173742).$$

$$\nabla_{e_1} \mathcal{L} = (-.347485, 0), \quad \nabla_{e_n} \mathcal{L} = (0, -.173742).$$

Scatter these into rows 0 and 4 of E ; every other row receives zero from this lookup.

Backward 5: split, scatter, and update only rows read

$$\nabla_{e_{\cdot}} \mathcal{L} = (-.347485, 0), \quad \nabla_{e_n} \mathcal{L} = (0, -.173742).$$

Scatter these into rows 0 and 4 of E ; every other row receives zero from this lookup. With $\eta = .1$:

$$e_{\cdot} \leftarrow (1.03475, 0), \quad e_n \leftarrow (0, 2.01737).$$

HELD CASE · OPENING · LEARNING The rows for $\cdot$ and n move. Target row a was not read and does not move through the lookup.

One scalar loss teaches every used block



One scalar loss teaches every used block



- output rows learn which hidden features support each token;
- hidden weights learn combinations of ordered context coordinates;
- read embedding rows learn features that improve future predictions.

The path is ordinary backpropagation; language structure entered when we built context-target pairs.

TRAIN repeats the calculation over a declared batch

TRAIN true context $\rightarrow \underline{p}$; CE $\rightarrow$ update

INFER current context $\rightarrow \underline{p}$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$\mathcal{L}_B = -\frac{1}{|B|} \sum_{i \in B} \log p_{\theta}(y_i | c_i).$$

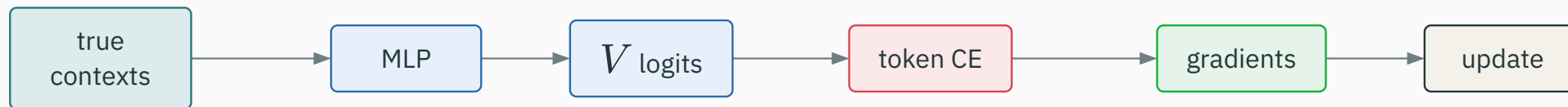
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EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$\mathcal{L}_B = -\frac{1}{|B|} \sum_{i \in B} \log p_{\theta}(y_i | c_i).$$



The notebook uses the entire training set as B on each Adam step: full-batch training, not a minibatch claim.

Split whole names before creating windows

TRAIN true context $\rightarrow \underline{p}$; CE $\rightarrow$ update

INFER current context $\rightarrow \underline{p}$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

TRAIN

10 whole names $\rightarrow$ 56 context-target pairs
weights receive updates

VALIDATION

4 different whole names $\rightarrow$ 30 pairs
no updates

Split whole names before creating windows

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10 whole names $\rightarrow$ 56 context-target pairs
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VALIDATION

4 different whole names $\rightarrow$ 30 pairs
no updates

Do not randomly split windows from the same name. Nearby windows share almost all their evidence.

**The split unit must match the object whose generalization we claim:
here, a whole name.**

EVAL declares tokenizer, targets, and reduction

TRAIN true context $\rightarrow \underline{p}$; CE $\rightarrow$ update

INFER current context $\rightarrow \underline{p}$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | c_i), \quad \text{PPL} = \exp(\text{NLL}).$$

EVAL declares tokenizer, targets, and reduction

TRAIN true context $\rightarrow \underline{p}$; CE $\rightarrow$ update

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$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | c_i), \quad \text{PPL} = \exp(\text{NLL}).$$

TOKENIZER

same character vocabulary and
boundary policy

TARGETS

N non-padding next-token targets

DATA

same held-out corpus and token-
weighted reduction

Perplexities are comparable only when all three contracts agree.

The executed notebook supplies held-out evidence

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

FINAL TRAIN

NLL = .645308
PPL = 1.90657

FINAL VALIDATION

NLL = .677965
PPL = 1.96986

The executed notebook supplies held-out evidence

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

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FINAL TRAIN

NLL = .645308
PPL = 1.90657

FINAL VALIDATION

NLL = .677965
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Same architecture class, widened in the notebook to $d = 4$, $H = 24$ and trained for 300 full-batch Adam steps.

The small train-validation gap says this tiny held-out set transfers similarly; it does not establish broad name quality.

Uniform prediction is the first evaluation sanity check

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

For $V = 6$, a uniform model has $p(y) = \frac{1}{6}$:

Uniform prediction is the first evaluation sanity check

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For $V = 6$, a uniform model has $p(y) = \frac{1}{6}$:

$$\text{NLL} = -\log\left(\frac{1}{6}\right) = \log 6 = 1.79176, \quad \text{PPL} = 6.$$

Uniform prediction is the first evaluation sanity check

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UNIFORM

1.79176 NLL
6.00 PPL

VALIDATION

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UNIFORM

1.79176 NLL
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VALIDATION

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The trained model predicts held-out next characters far better than uniform guessing under this protocol.

The same p now crosses from TRAIN to INFER

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

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EVAL held-out truth $\rightarrow$ mean NLL / PPL

TRAIN

true context $\rightarrow$ distribution $\rightarrow$ compare with true target
 $\rightarrow$ update

INFER

current context $\rightarrow$ distribution $\rightarrow$ choose token $\rightarrow$
append

The same p now crosses from TRAIN to INFER

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

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EVAL held-out truth $\rightarrow$ mean NLL / PPL

TRAIN

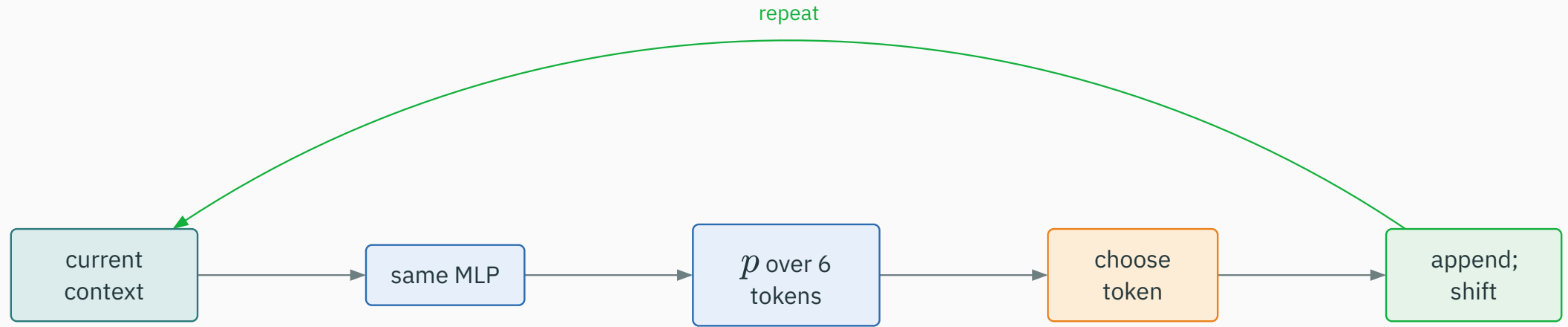
true context $\rightarrow$ distribution $\rightarrow$ compare with true target
 $\rightarrow$ update

INFER

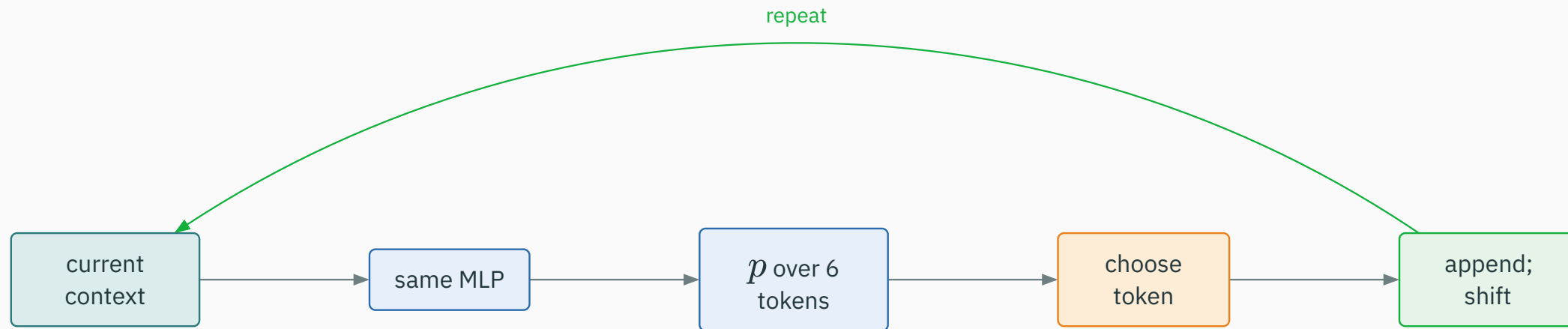
current context $\rightarrow$ distribution $\rightarrow$ choose token $\rightarrow$
append

Weights stay fixed during generation; the chosen token changes the next input context.

Autoregressive generation closes a feedback loop

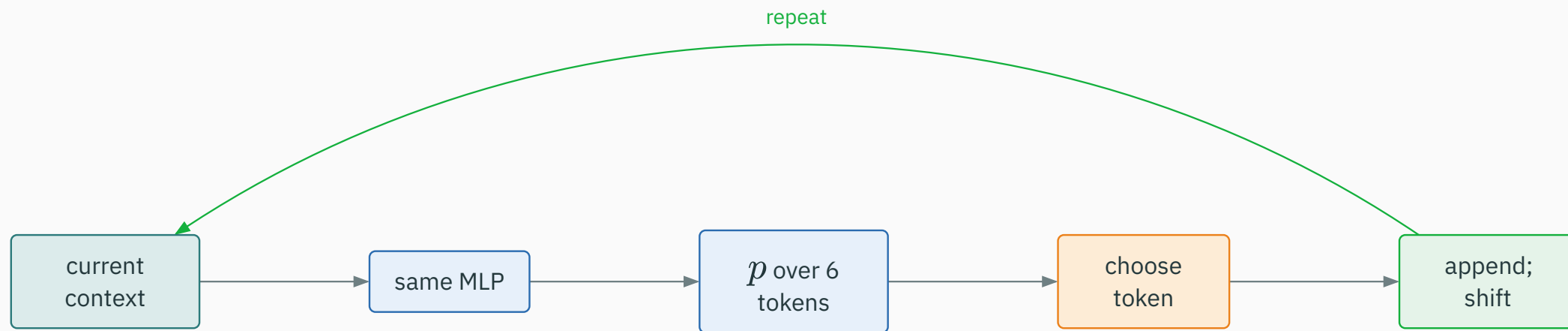


Autoregressive generation closes a feedback loop



Start from . . ; stop when the model emits . or reaches a declared maximum length.

Autoregressive generation closes a feedback loop

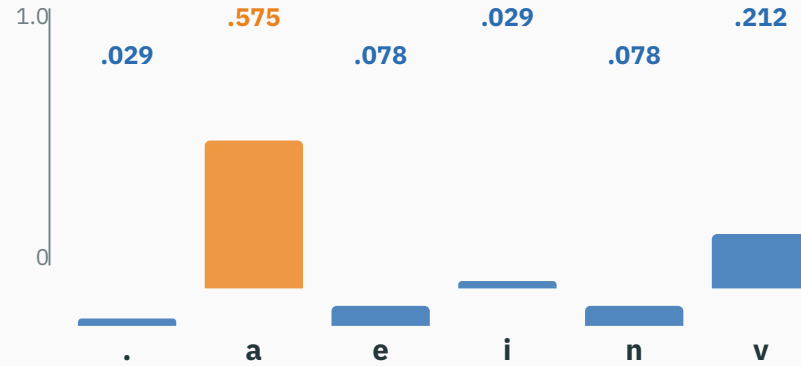


Start from . . ; stop when the model emits . or reaches a declared maximum length.

Autoregressive means the model conditions on tokens already present, including its own earlier choices.

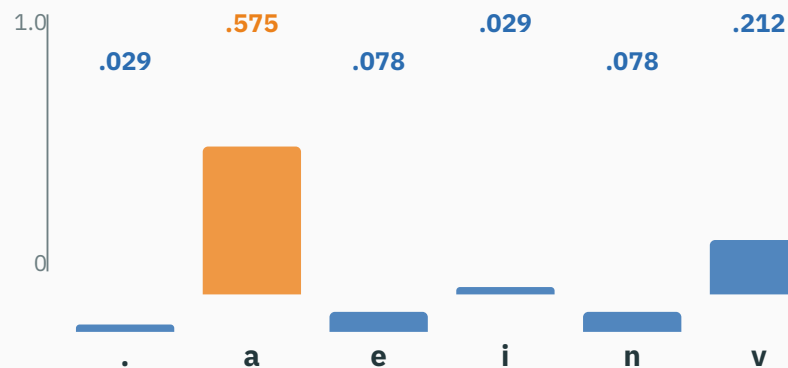
Commit: must greedy and sampling choose the same token?

QUESTION



Commit: must greedy and sampling choose the same token?

QUESTION



GREEDY

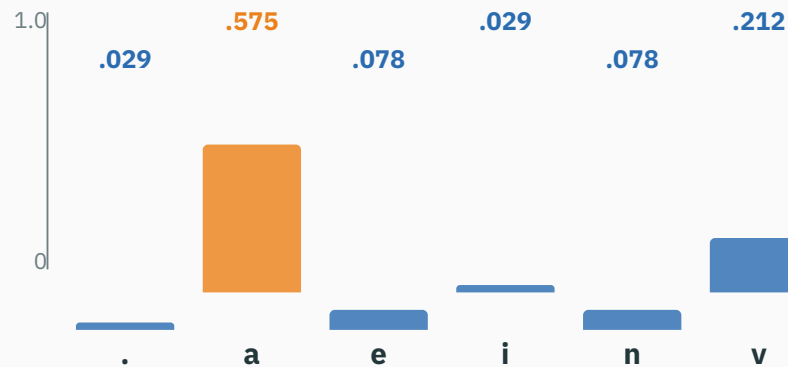
$$\hat{x} = \arg \max_v p(v|.n)$$

SAMPLING

$$\hat{x} \sim \text{Categorical}(p)$$

Commit: must greedy and sampling choose the same token?

QUESTION



GREEDY

$$\hat{x} = \arg \max_v p(v|.n)$$

SAMPLING

$$\hat{x} \sim \text{Categorical}(p)$$

No. Greedy always chooses a; sampling can choose any token with nonzero mass.

One uniform draw makes categorical sampling concrete

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

token	.	a	e	i	n	v
<i>p</i>	.029	.575	.078	.029	.078	.212
CDF	.029	.604	.682	.710	.788	1.000

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Draw $u = .650$ from $\text{Uniform}(0, 1)$.

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$.604 < .650 \leq .682 \Rightarrow$ **sample e.**

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Draw $u = .650$ from $\text{Uniform}(0, 1)$.

$.604 < .650 \leq .682 \Rightarrow$ **sample e.**

The model supplies probabilities; the decoder supplies the decision rule and randomness.

Temperature rescales logits before the same softmax

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$p_{i(\tau)} = \text{softmax}\left(\frac{z}{\tau}\right)_i, \quad \tau > 0.$$

Temperature rescales logits before the same softmax

D DERIVATION

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

$$p_{i(\tau)} = \text{softmax}\left(\frac{z}{\tau}\right)_i, \quad \tau > 0.$$

$$\tau = .5$$

logit gaps double
distribution concentrates

$$\tau = 1$$

learned distribution
no rescaling

$$\tau = 2$$

logit gaps halve
distribution flattens

Temperature rescales logits before the same softmax

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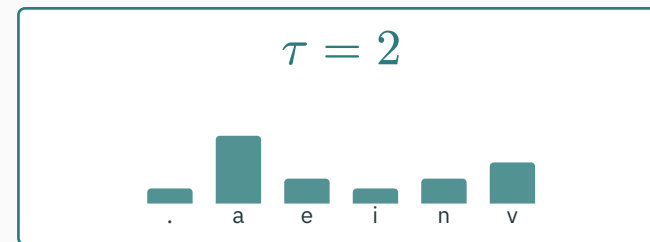
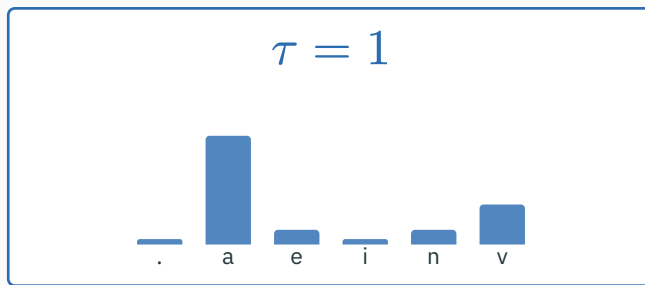
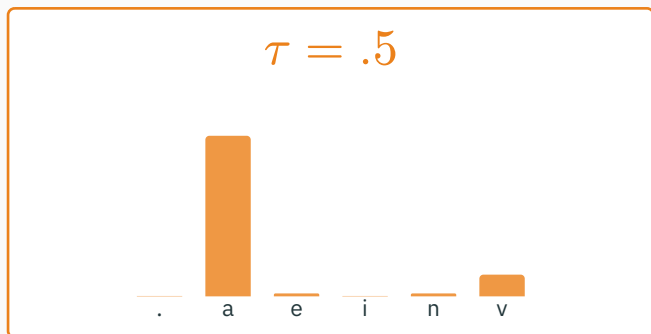
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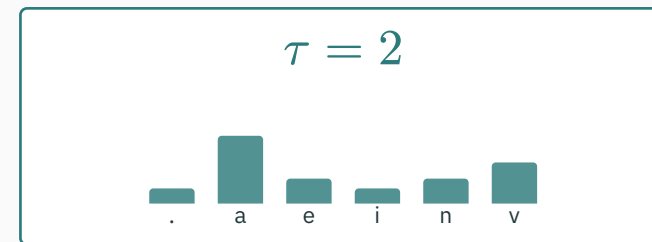
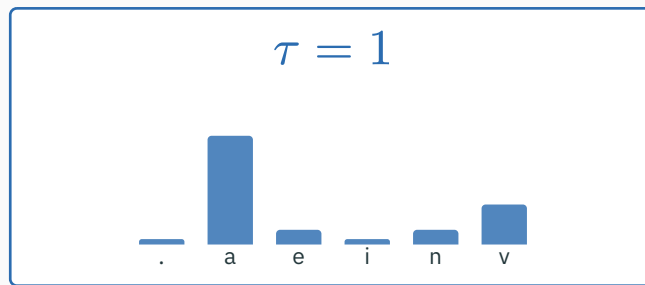
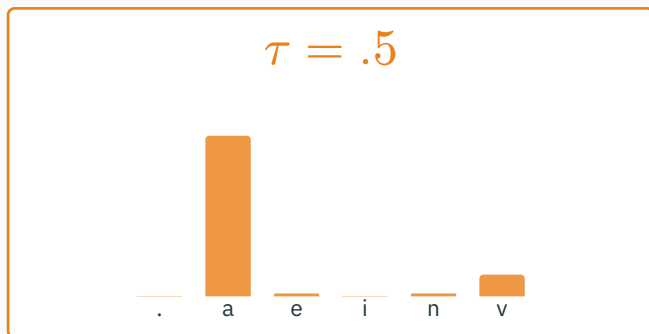
logit gaps halve
distribution flattens

Temperature changes neither weights nor token ranking for positive τ .

The same six logits produce three exact distributions

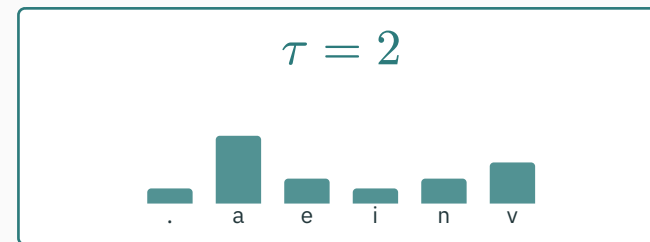
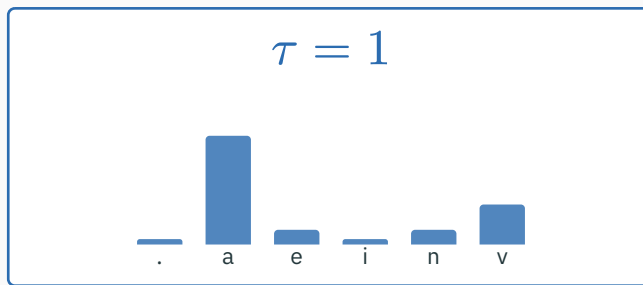
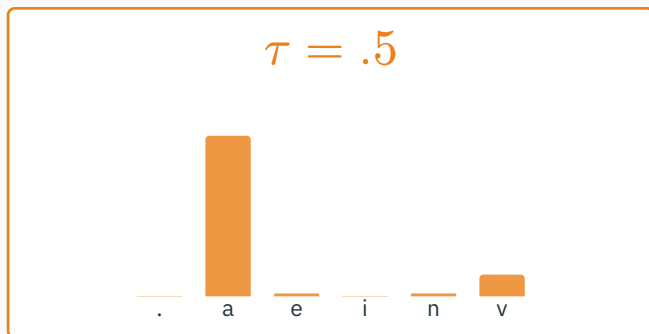


The same six logits produce three exact distributions



τ	.	a	e	i	n	v
.5	.002	.850	.016	.002	.016	.115
1	.029	.575	.078	.029	.078	.212
2	.080	.359	.132	.080	.132	.218

The same six logits produce three exact distributions



τ	.	a	e	i	n	v
.5	.002	.850	.016	.002	.016	.115
1	.029	.575	.078	.029	.078	.212
2	.080	.359	.132	.080	.132	.218

Concentration is measurable; “safer” or “more correct” is not guaranteed.

Choose temperature by behavior, not a moral label

QUESTION

SYMPTOM

samples collapse to the same spelling

TEST

hold logits and seed fixed; compare token frequencies across

$\mathcal{T}$

Choose temperature by behavior, not a moral label

QUESTION

SYMPTOM

samples collapse to the same spelling

TEST

hold logits and seed fixed; compare token frequencies across

τ

LOWER τ

more mass on current top tokens; less variation

HIGHER τ

more mass on alternatives; more variation

Choose temperature by behavior, not a moral label

QUESTION

SYMPTOM

samples collapse to the same spelling

TEST

hold logits and seed fixed; compare token frequencies across

τ

LOWER τ

more mass on current top tokens; less variation

HIGHER τ

more mass on alternatives; more variation

If the model ranks a bad token highest, lowering temperature can make that error more consistent.

The trained MLP is blind to prefixes with the same suffix

TRAIN true context $\rightarrow p$; CE $\rightarrow$ update

INFER current context $\rightarrow p$; choose $\rightarrow$
append

EVAL held-out truth $\rightarrow$ mean NLL / PPL

PREFIX A

na $\rightarrow$ visible context na

$\equiv$

PREFIX B

veena $\rightarrow$ visible context na

The trained MLP is blind to prefixes with the same suffix

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$$p = (.418671, 9.91 \times 10^{-7}, 3.77 \times 10^{-6}, .142728, 3.17 \times 10^{-5}, .438565)$$

The trained MLP is blind to prefixes with the same suffix

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$$p = (.418671, 9.91 \times 10^{-7}, 3.77 \times 10^{-6}, .142728, 3.17 \times 10^{-5}, .438565)$$

The two distributions are exactly identical because the model receives exactly the same two ids.

Diagnose the architecture before tuning the optimizer

SYMPTOM

different long prefixes produce identical
next-token distributions

SUSPECT

the useful distinction lies outside the
fixed window

TEST

hold the suffix fixed, change only earlier
tokens, compare p exactly

SYMPTOM

different long prefixes produce identical
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If p cannot change, more training cannot make this architecture use the hidden prefix.

SYMPTOM

different long prefixes produce identical
next-token distributions

SUSPECT

the useful distinction lies outside the
fixed window

TEST

hold the suffix fixed, change only earlier
tokens, compare p exactly

If p cannot change, more training cannot make this architecture use the hidden prefix.

This is an information-access failure, not evidence that Adam, embeddings, or temperature are broken.

A larger window delays the failure but does not remove it

D DERIVATION

$W_h \in \mathbb{R}^{H \times kd} \Rightarrow Hkd \text{ weights.}$

A larger window delays the failure but does not remove it

$W_h \in \mathbb{R}^{H \times kd} \Rightarrow Hkd \text{ weights.}$

- doubling k doubles this block;
- the same token in slot 1 and slot 2 meets different columns;
- the maximum usable history remains fixed at model construction.

A larger window delays the failure but does not remove it

$W_h \in \mathbb{R}^{H \times kd} \Rightarrow Hkd \text{ weights.}$

- doubling k doubles this block;
- the same token in slot 1 and slot 2 meets different columns;
- the maximum usable history remains fixed at model construction.

Concatenation preserves order, but the MLP does not reuse one position-independent update across time.

Revisit: every opening commitment now has evidence

QUESTION

HELD CASE · OPENING REVISIT context .n, exact six-token model.

OUTPUT

$$p = (.029, .575, .078, .029, .078, .212)$$

CHOICE

greedy $\rightarrow$ a; sample with $u = .65$
 $\rightarrow$ e

LEARNING

lookup updates rows . and n

Revisit: every opening commitment now has evidence

QUESTION

HELD CASE · OPENING REVISIT context .n, exact six-token model.

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$$p = (.029, .575, .078, .029, .078, .212)$$

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greedy $\rightarrow$ a; sample with $u = .65$
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LEARNING

lookup updates rows . and n

One distribution supports training, decoding, and evaluation only after the clock and policy are declared.

OPTIONAL: expose the full output-gradient matrix

With $h = (1, 0)$,

$$\nabla_{w_o} \mathcal{L} = \begin{pmatrix} .028644 & 0 \\ -.424667 & 0 \\ .077863 & 0 \\ .028644 & 0 \\ .077863 & 0 \\ .211653 & 0 \end{pmatrix}.$$

OPTIONAL: expose the full output-gradient matrix

With $h = (1, 0)$,

$$\nabla_{w_o} \mathcal{L} = \begin{pmatrix} .028644 & 0 \\ -.424667 & 0 \\ .077863 & 0 \\ .028644 & 0 \\ .077863 & 0 \\ .211653 & 0 \end{pmatrix}.$$

Each vocabulary row receives its own logit error; the inactive hidden feature leaves the second column zero.

OPTIONAL: use the notebook as a prediction protocol

I INTERACTIVE

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L11/01_char_mlp_language_model.ipynb

Before running: predict the seven windows, active ReLU unit, sign of the target gradient, two embedding rows that move, and whether `na` and `veena` can differ.

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VERIFY

exact forward/backward arrays; train/validation NLL;
generated names

EXPLAIN

why temperature changes diversity and why a shared suffix
forces identical p

OPTIONAL: use the notebook as a prediction protocol

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generated names

EXPLAIN

why temperature changes diversity and why a shared suffix
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Prediction before execution turns the notebook from a demo into evidence.

PRIMARY SOURCE

Bengio et al. (2003)
distributed representations plus a feedforward neural
probabilistic language model

FOLLOW-ALONG

Exact ES 667 Colab
deterministic NumPy forward/backward, full-batch training,
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generation, and suffix test

The paper scales the central idea; our six-token model keeps its mechanics auditable by hand.

Next: replace a fixed window with a shared recurrent state

THIS LECTURE

$c_t = [e_{t-2}; e_{t-1}]$
fixed access; position-specific columns



PUBLIC L13

$h_t = f_{\theta}(h_{t-1}, e_t)$
shared update; variable-length stream

THIS LECTURE

$$c_t = [e_{t-2}; e_{t-1}]$$

fixed access; position-specific columns



PUBLIC L13

$$h_t = f_{\theta}(h_{t-1}, e_t)$$

shared update; variable-length stream

The next-token objective stays; the mechanism for carrying context changes.

A language model estimates a distribution over the **next token**.

Next: **RNNs, LSTMs, and GRUs** - learn what recurrent state remembers, forgets, and struggles to train.